Machine Learning Applications for Health

COMP90089

Instructor Info

Prof Daniel Capurro, PhD and Dr. Vlada Rozova

Office Hrs: By appointment

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Course Info

Tuesdays

3:00am-5:00pm

Q230 Theatre - Kwong Lee Dow Building - Level 2

Tutor Info

Roben Delos Reyes

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Overview

Artificial Intelligence (AI) is an ever growing field that holds the promise to revolutionise the way we develop drugs, treat and manage patients. In particular, Machine Learning has become an important tool to make sense of clinical data that is routinely collected and stored as electronic health records to enable personalised medicine.

This subject aims to introduce students to different Machine Learning applications in health, using different clinical data sources and computational techniques, discussing their idiosyncrasies and the main challenges in the area.

Material

Required Texts

There are no required text books on this subject. Any required journal articles, book chapters, or other reading material will be provided on Canvas. Videos will also be provided on Canvas.

Assessment

Note that you must achieve at least a 50% pass in *each of the individual and group assignments* in order to pass the subject.

20%	Regular, in-class (weekly) quizzes - Equivalent to 500 words in total, from week 2 to week 11
30%	Two individual assignments - Equivalent to 2000 words (2x1000 words). These will be individual programming assignments about one of the Machine Learning applications in health presented during the subject. Due in Week 4 & 10.
10%	AI in health project proposal (group activity, 4 students, oral presentation) - Equivalent to 500 words per group. Due in Week 8.
40%	AI in health project written report (20%) + oral presentation (20%)(group activity, 4 students) - Equivalent to 2000 words per group. Due at the start of the exam period.

Every assignment will be checked for originality and the expectation is that every work will be:

- Original work done by the author(s) of the assignment *during* this subject, *during* this semester;
- Appropriately referenced (meaning every claim needs a reference; references will be checked). You can pick the citation style but please be consistent.
- Generative AI can be used for brainstorming, planning, proofreading and editing (as described here) but *should not be used for drafting text or content generation*. Its use should be disclosed as per the University of Melbourne's current policy. Every written/code assignment should have a statement by the student or group of students stating:
  - Whether a generative AI was used
  - Which tool was used
  - The purpose it was used for (brainstorming, planning, proofreading, or editing)
  - A declaration that it was not used for drafting text/code or content generation.

In case the work does not meet these expectations, the assignment will penalised with a 30% discount on the mark. A late penalty of 5% per unjustified day of delay also applies.

Curving of grades is at the discretion of the course instructor.

FAQs

Can I pick my own group for the project?

No, your groups will be assigned by the instructor so they are diverse enough to enhance peer learning.

Can I pick my own topic for the project?

Yes, but you need to agree as a group what your topic will be.

Can I do my project individually?

No

What can I do if my group is not working for me?

Refer to the group contract you signed on week 1 and please let the instructor know as soon as possible.

Learning Objectives

On completion of this subject, students should be able to:

- Describe the application of AI concepts in the context of health problems
- Demonstrate familiarity with challenges associated with health data and data modelling
- Design, implement & evaluate an AI system addressing a healthcare problem
- Critically assess Artificial Intelligence applications for health

Generic Skills:

- Develop critical thinking and analytical skills
- Improve problem-solving skills
- Enhance programming skills
- Improve communication skills
- Develop skills to enable problem identification, solution formulation for health applications

Subject Communication

Announcements will be posted on Ed Discussion, please make sure you review the platform and your student email address.

If you need to communicate with the instructors or tutors, please use Ed Discussion. If you want to send an email, use the following text in the email's subject: [COMP90089 inquiry].

If you have questions about the subject, assessments, etc, please post them at the course's forum so everyone can see the question and the answers. This forum is also good for establishing student-to-student interactions.

Diversity and Inclusivity Statement

I consider this course to be a place where you will be treated with respect, and I welcome individuals of all ages, backgrounds, beliefs, ethnicities, genders, gender identities, gender expressions, national origins, religious affiliations, sexual orientations, ability - and other visible and non-visible differences. All members of this class are expected to contribute to a respectful, welcoming and inclusive environment for every other member of the class.

Student Support Services

The University offers many support services for students.

Some of the key services are listed below (links are clickable):

- Library
- Academic Skills
- Student Equity and Disability Support
- Student wellbeing
- Special consideration processes
- Stop 1 Support
- Career support

Academic Integrity

The University has a Student Academic Integrity Policy. Students are expected to be familiar with the code and to understand that every submitted assignment is a product of their own original work and faithfully represents the amount of effort and time invested in producing it. Every use of external work should be adequately referenced. Violations of the policy will be handled according to the procedures described in the aforementioned policy.

You can also check the Academic Integrity Website. Additional information is available in the Academic Integrity Module inside the course's Canvas site.

Class Schedule

MODULE 1: Introduction

Week-1 28/07/26	Lecture #1: Course Introduction & Housekeeping	Prof Daniel Capurro
	Lecture #2: Biomedical Informatics & the Learning Healthcare System	Prof Daniel Capurro
Week-2 04/08/26	Lecture #3: Data Sources	Prof Daniel Capurro
	Lecture #4: Clinical Data Types	Dr Vlada Rozova
Week-3 11/08/26	Lecture #5: Privacy, De-identification, Ethics, and Governance	Dr. Vlada Rozova
	Lecture #6: Deep Dive into Clinical Data	Prof Daniel Capurro

MODULE 2: Why is healthcare AI unique?

Week-4 18/08/26	Lecture #7: Digital Phenotyping	Prof Daniel Capurro
	Lecture #8: Digital Phenotyping Applications	Prof Daniel Capurro
Week-5 25/08/26	Lecture #9: Diagnostic Reasoning (I)	Prof Daniel Capurro
	Lecture #10: Diagnostic Reasoning (II)	Prof Daniel Capurro
Week-6 01/09/26	Lecture #11: Defining the Clinical Task	Prof Daniel Capurro
	Lecture #12: Study Design	Dr. Vlada Rozova
Week-7 08/09/26	Lecture #13: Clinical Accuracy Metrics	Dr. Vlada Rozova
	Lecture #14: AI Lifecycle	Dr. Vlada Rozova
Week-8 15/09/26	Lecture #15: Implementation and clinical workflows	Prof Daniel Capurro

MODULE 3: Main techniques

Week-8 15/09/26	Lecture #16: Deep Learning	Dr. Vlada Rozova
Week-9 22/09/26	Lecture #17: Clinical NLP	A/Prof Mike Conway
	Lecture #18: Large Language Models	Dr. Vlada Rozova
Week-10 06/10/26	Lecture #19: AI Agents	Prof Daniel Capurro
	Lecture #20: Combining Methods	Prof Daniel Capurro

MODULE 4: Implementation & Legal aspects

Week-11 13/10/26	Lecture #21: Evidence to decision framework	Dr. Vlada Rozova
	Lecture #22: Legal/Regulatory aspects of ML Applications	TBD
Week-12 20/10/26	Lecture #23: Project Discussion	Prof Daniel Capurro
	Lecture #24: Project Discussion	Prof Daniel Capurro
Week-14 03/11/26	Lecture #25: Project Presentations – Specific Schedule TBD, will happen during lecture and tutorial hours	All